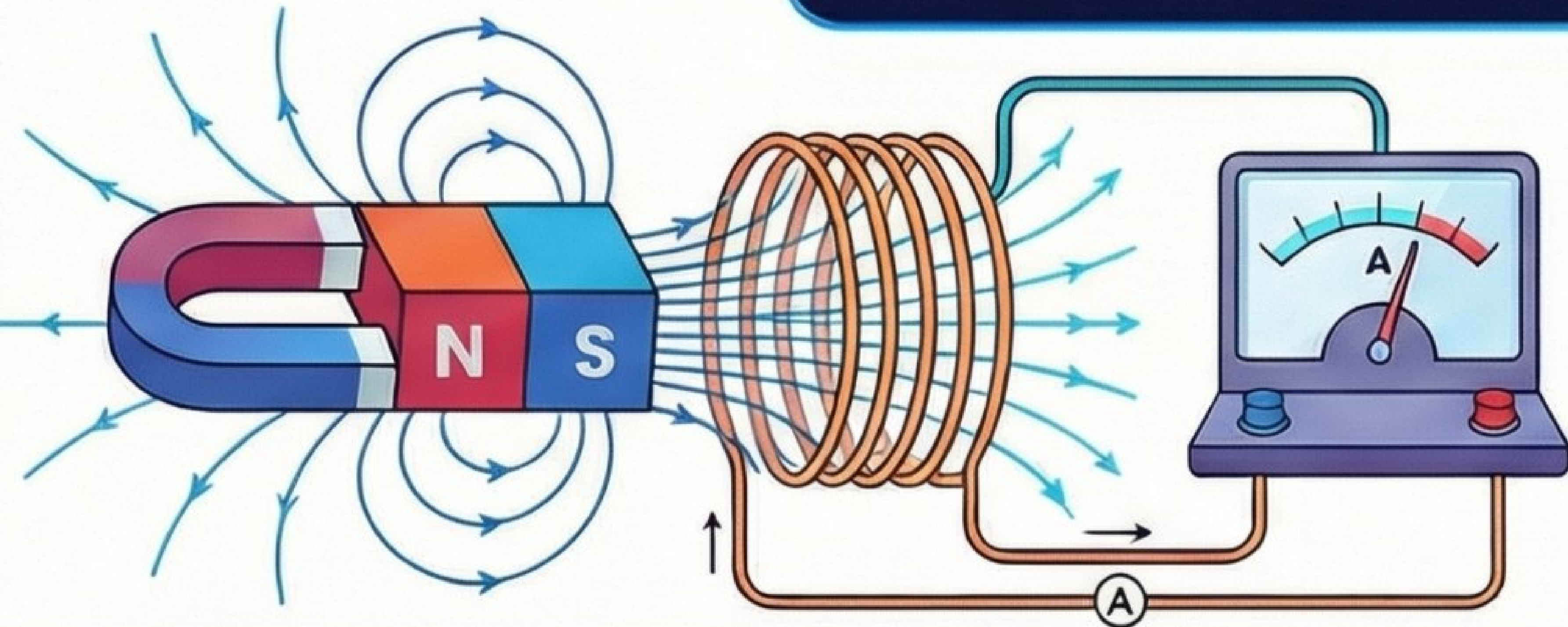


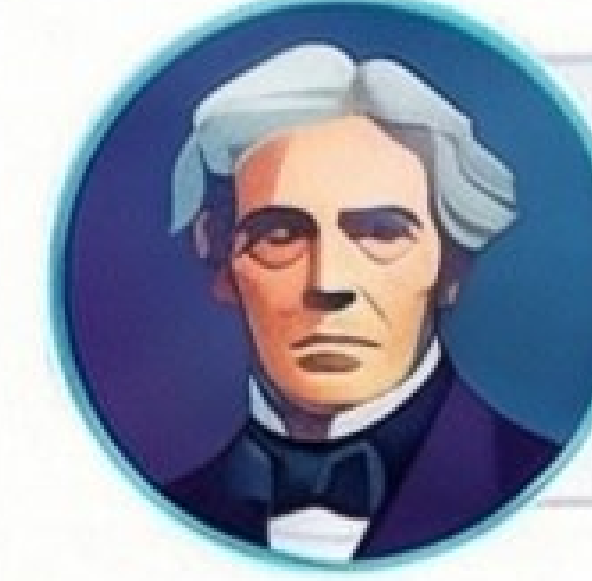
A Quick Guide to Electromagnetic Induction



THE FOUNDATIONS OF EMI

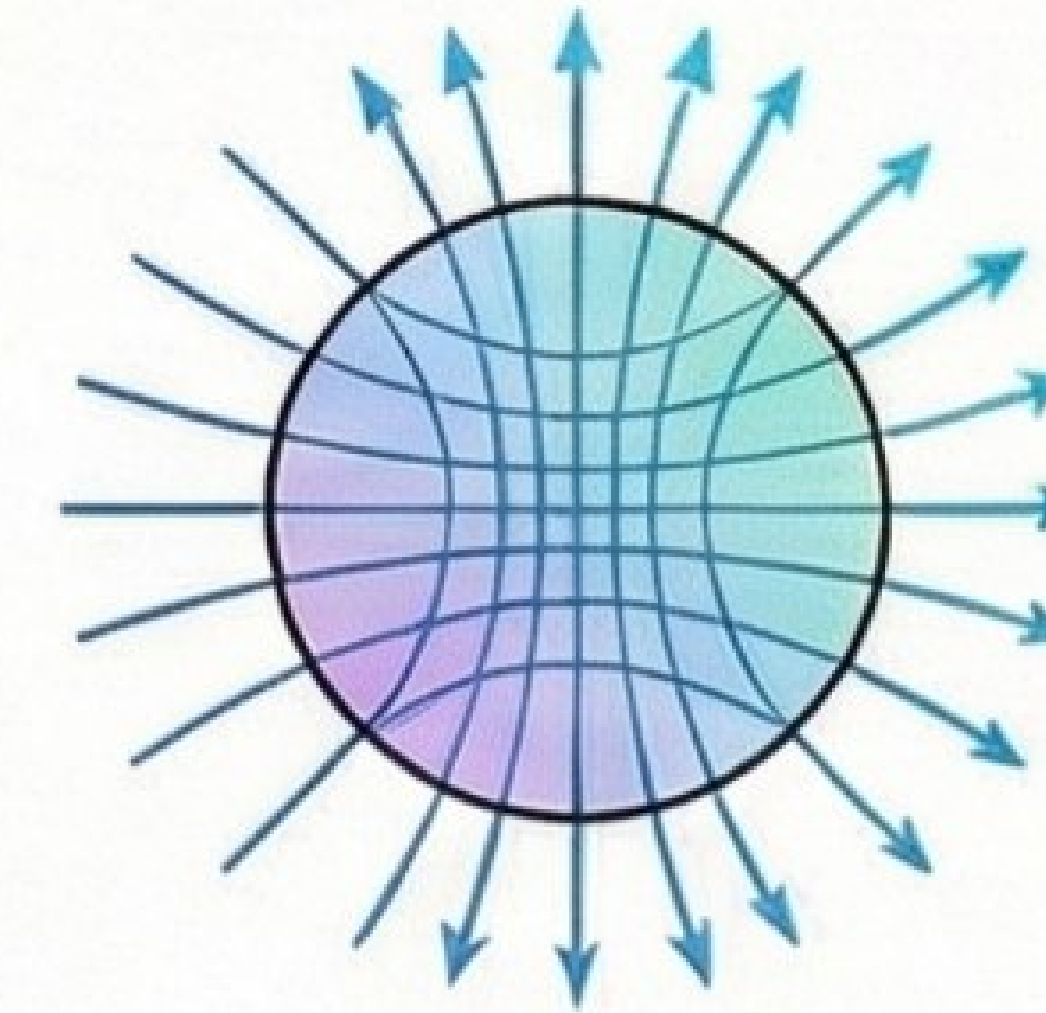
What is Electromagnetic Induction?

The process of producing an electric current (or EMF) in a coil by varying the magnetic field linked with it.



Discovered by Michael Faraday

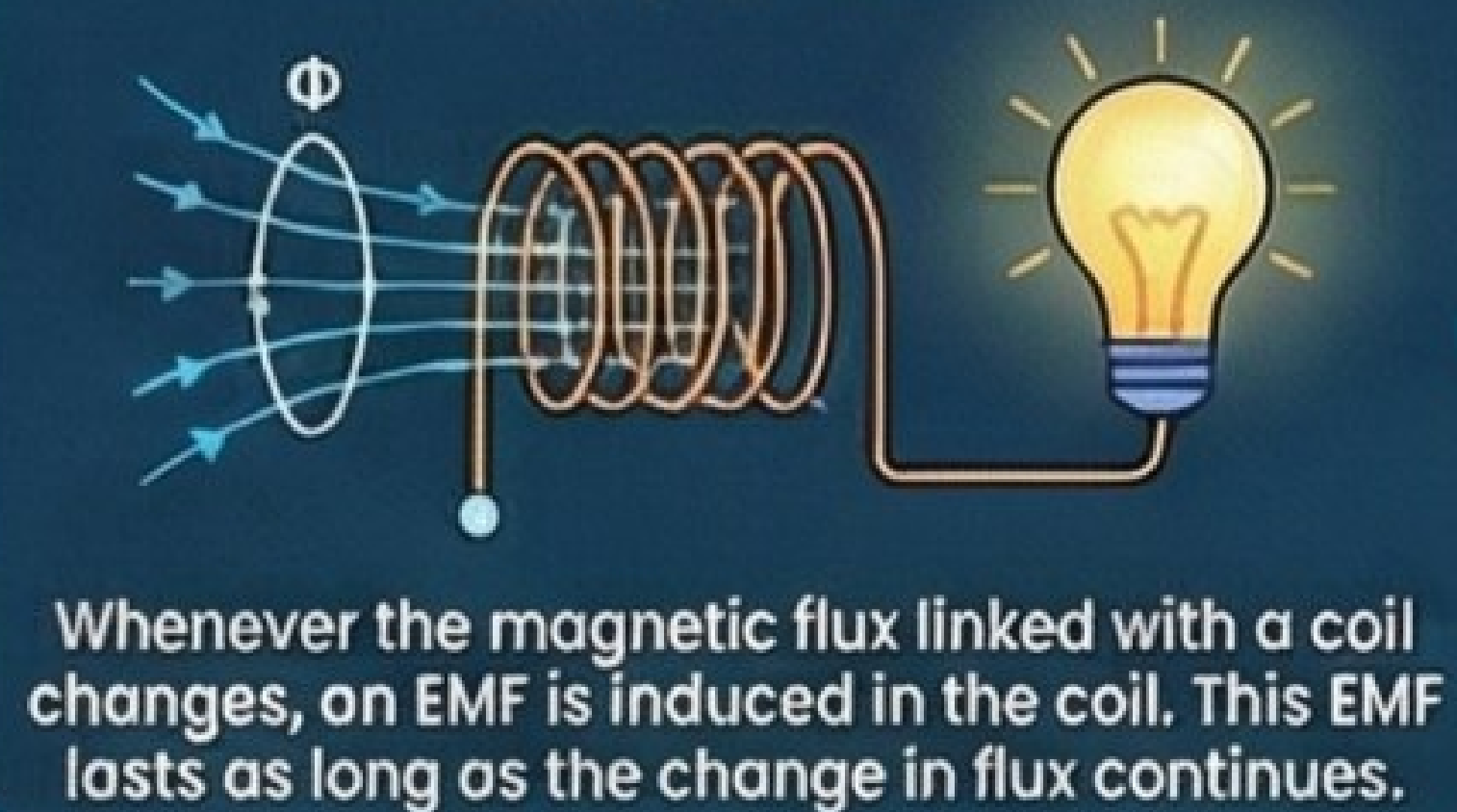
Faraday demonstrated that moving a magnet near a coil, or vice-versa, caused a galvanometer to show a deflection, indicating a current.



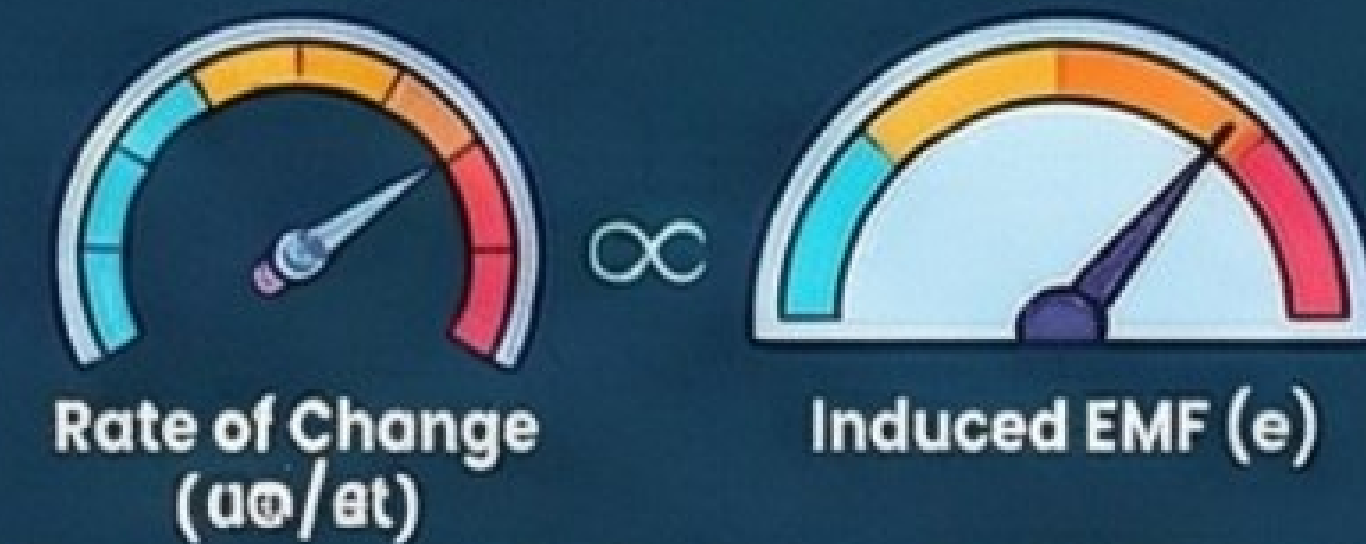
The Role of Magnetic Flux (Φ)

Magnetic flux is the measure of the total magnetic field lines passing through a given area. EMI occurs only when this flux changes.

Faraday's First Law



Faraday's Second Law



The magnitude of the induced EMF is directly proportional to the rate of change of the magnetic flux. Faster change = more EMF.

FARADAY'S & LENZ'S LAWS

$$e = - \frac{d\Phi}{dt}$$

The induced EMF (e) equals the rate of change of magnetic flux ($d\Phi/dt$). The negative sign is explained by Lenz's Law.

Lenz's Law: The Law of Opposition

The direction of the induced current is always such that it opposes the very change in magnetic flux that produced it.

North Pole Approaching Coil



The coil's near face becomes a North pole to repel the magnet. This is achieved by an anti-clockwise current flow.

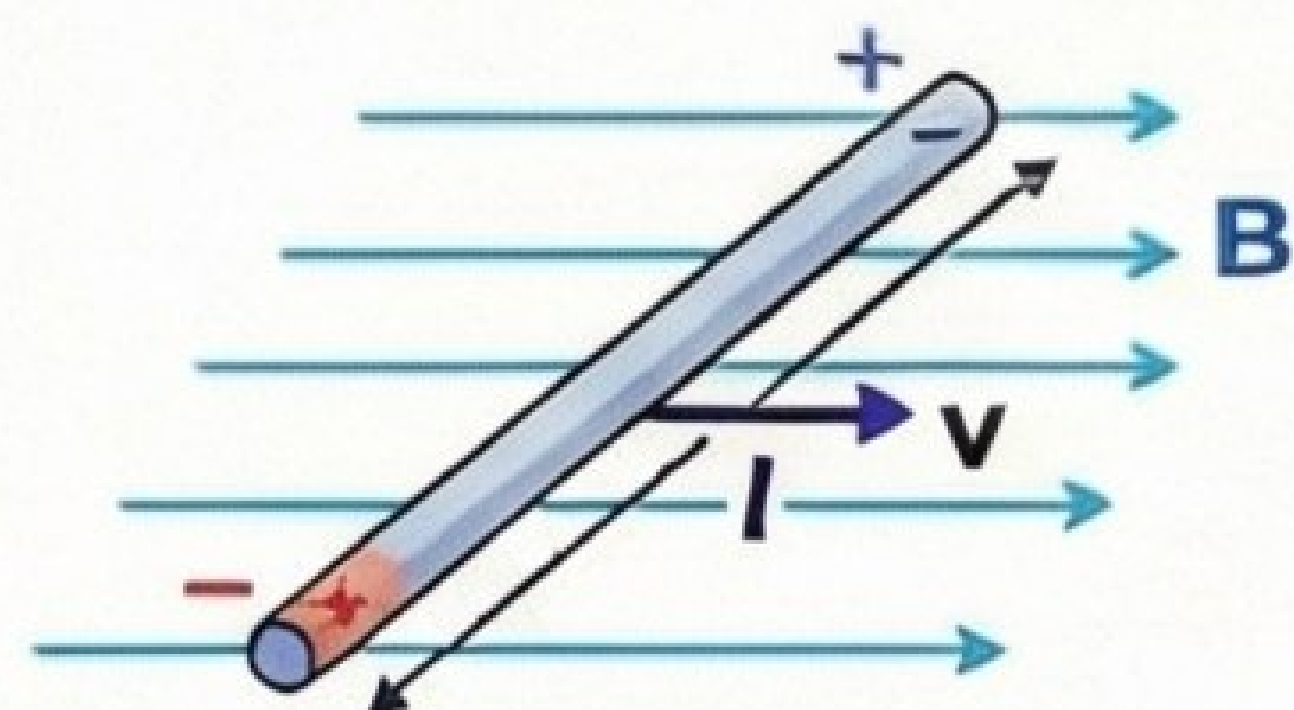
North Pole Moving Away



The coil's near face becomes a South pole to attract the magnet. This is achieved by a clockwise current flow.

KEY PHENOMENA & FORMULAS

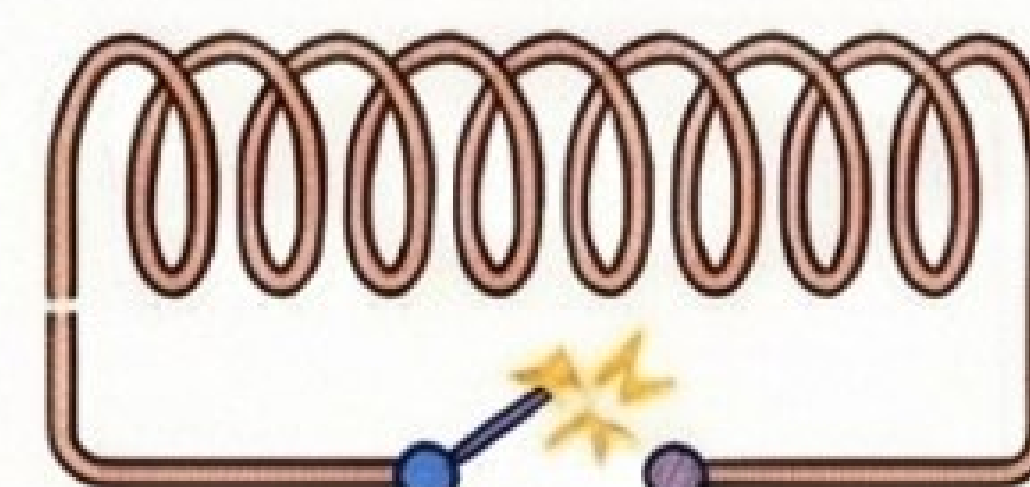
Motional EMF



$$e = Bvl$$

B = magnetic field, v = velocity, l = length

Self-Induction



Self-Inductance of a Solenoid:

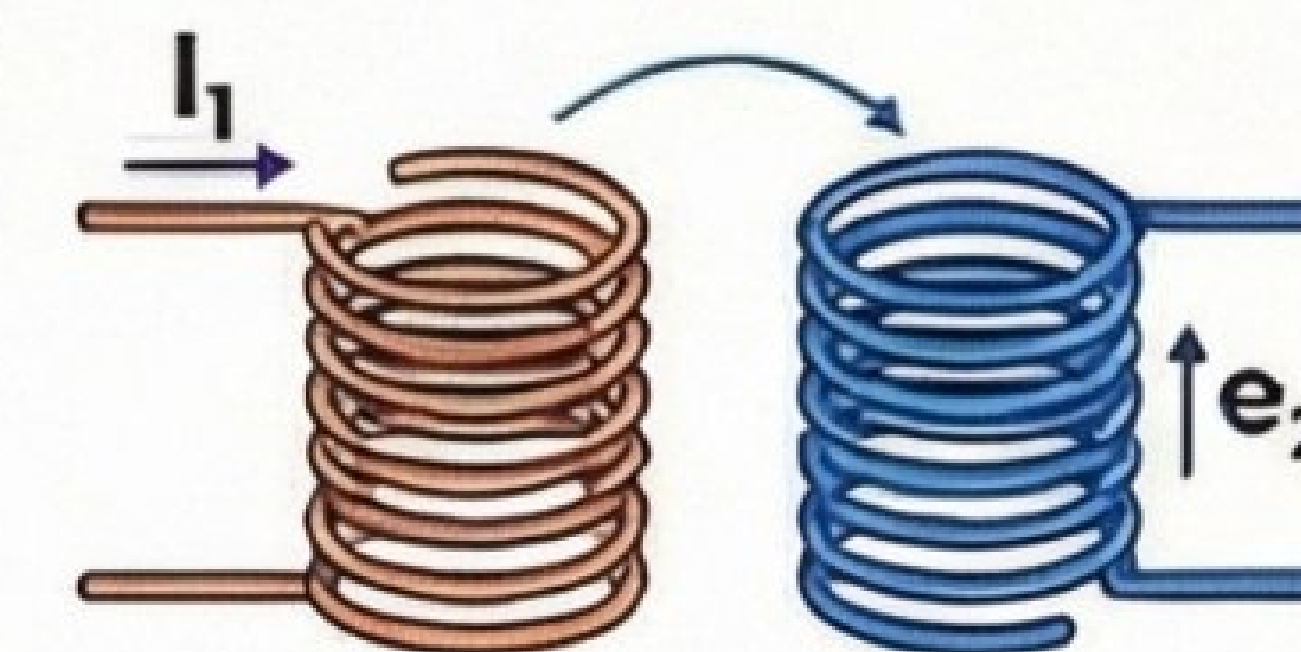
$$L = \mu_0 n^2 A l$$

Self-Inductance of a Solenoid

L = coefficient of self-inductance
 n = turn density
 A = area, l = length

This is why sparks can occur when a circuit is suddenly broken.

Mutual Induction



$$M = \mu_0 n_1 n_2 A l$$

The phenomenon where a changing current in one coil (primary) induces an EMF in a neighboring coil (secondary).

M

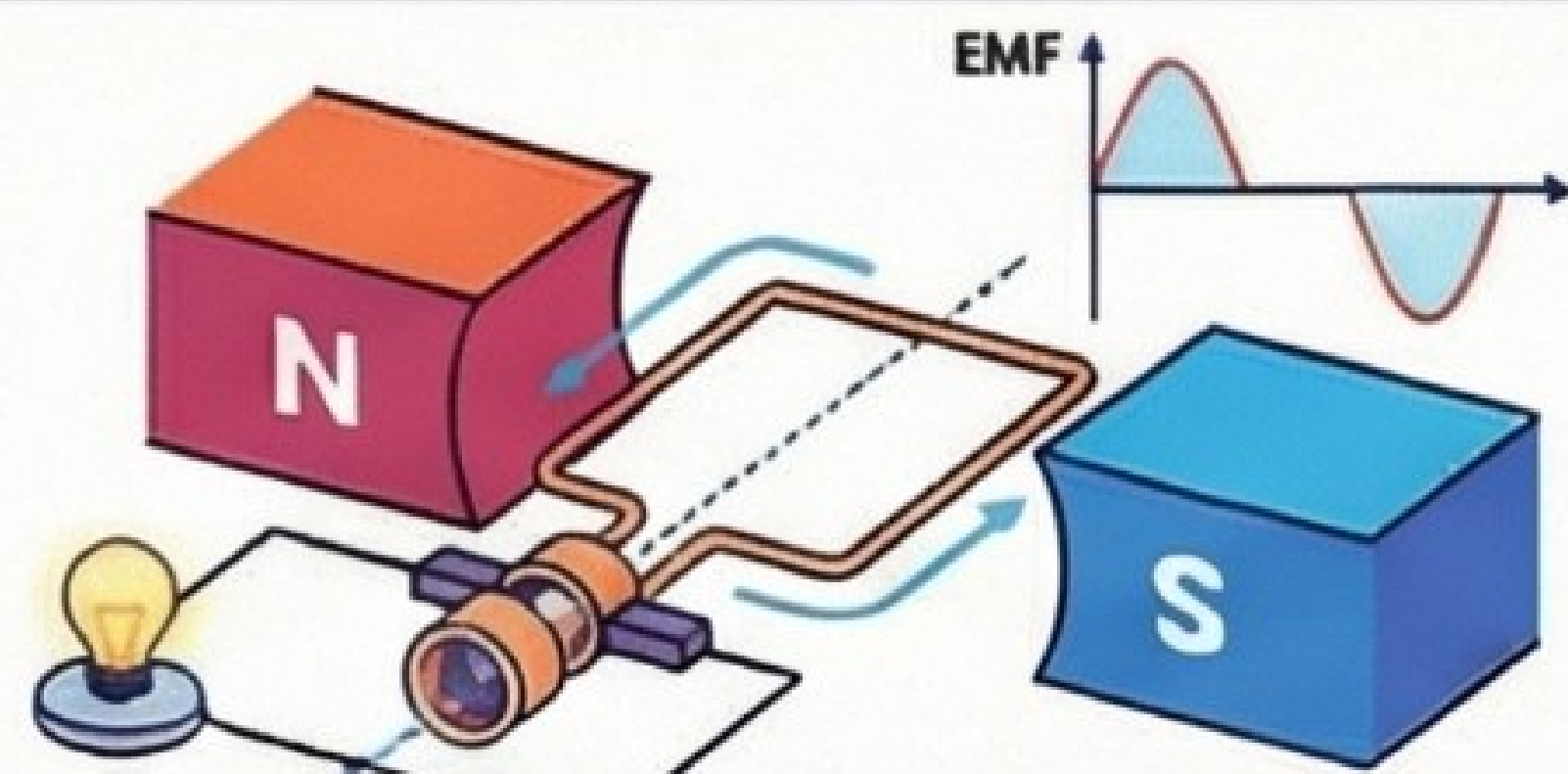
' M ' is the coefficient of mutual inductance, dependent on the properties of both coils.

AC Generator

A device that converts mechanical energy into alternating electrical energy based on the principle of electromagnetic induction.

How it Works

A coil is rotated within a uniform magnetic field. As it rotates, the magnetic flux through it changes continuously, inducing an alternating current.



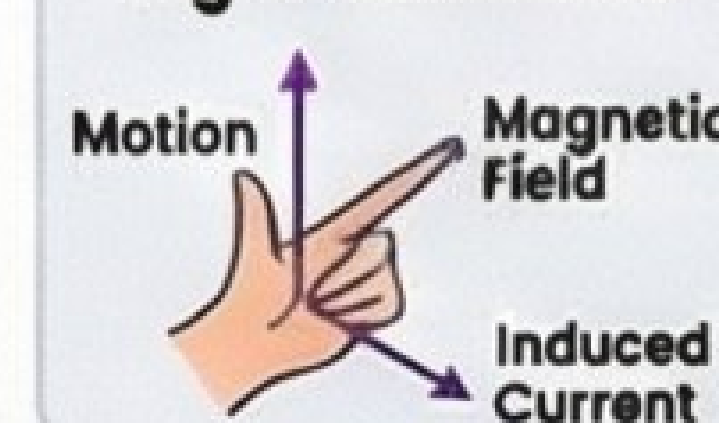
APPLICATIONS OF EMI

Induced EMF in an AC Generator:

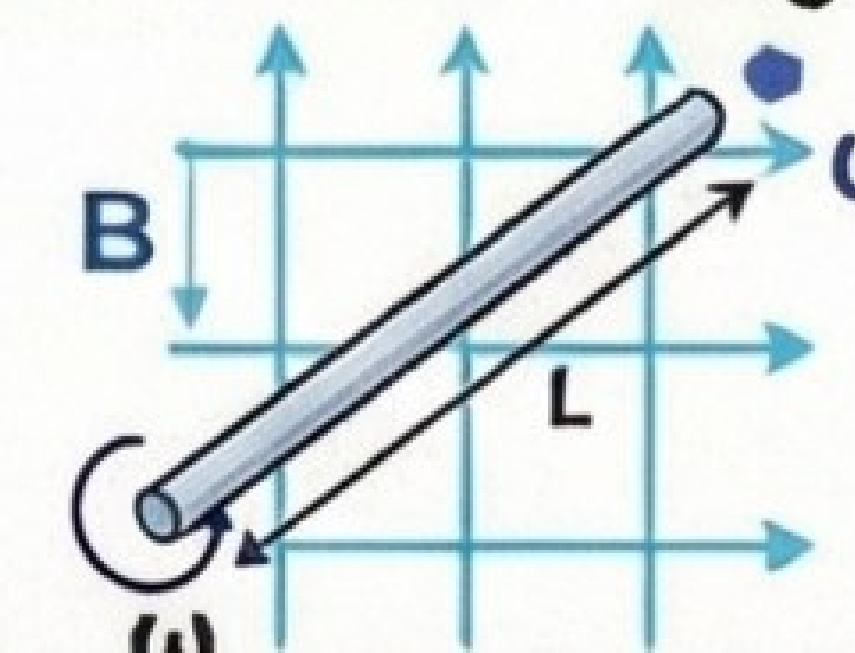
$$e = NBA\omega \sin(\omega t)$$

The EMF (e) varies sinusoidally with time.
The peak EMF is $e_0 = NBA\omega$.

Fleming's Right-Hand Rule



EMF in a Rotating Rod



Rotating Rod EMF:

$$e = \frac{1}{2} B\omega L^2$$

B = magnetic field
 ω = angular velocity
 L = length of the rod